

Non-Cirrhotic Diseases

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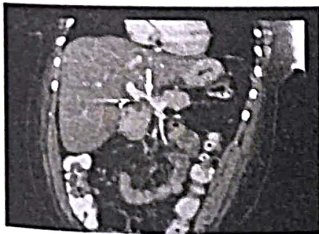
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EHPVO VS NCPF / IDIOPATHIC PORTAL HYPERTENSION (IPH)

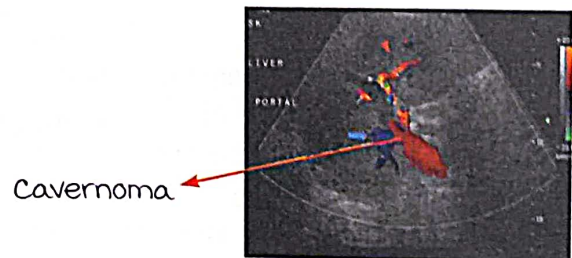
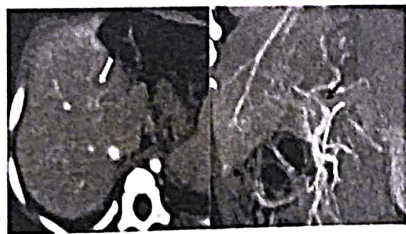
	EHPVO	NCPF/IPH/Hepatoportal Sclerosis Obliterative portal venopathy
Outcome	3-8 yr child (m/c cause of UGI bleed in children)	20-40 yr adult asian male.
Involvement	1st order portal vein	3rd order portal vein
Cause	Umbilical sepsis > prothrombotic state	E. coli/Arsenic associations, Hypervitaminosis A
Presentation	massive UGI bleed + massive spleen (hypersplenism) → Anemia, thrombocytopenia.	massive UGI bleed + massive spleen
Prognosis	Good, with no risk for cirrhosis	Good, negligible risk for cirrhosis
Investigations	IOC : CT • Cavernoma appearance of collaterals	• IOC : Doppler • Biopsy (Gold std) : Obliterative portal venopathy.
Complications	• Stunted growth • Portal biliopathy : compression of biliary system by portal collaterals.	—
Treatment	Endoscopic variceal ligation/Endoscopic sclerotherapy	—

Note :

m/c cause of PHTN : —→ Asia: EHPVO
—→ world: Biliary atresia



CECT: EHPVO showing collaterals



Color doppler : EHPVO



Sclerotic & non patent portal tracts

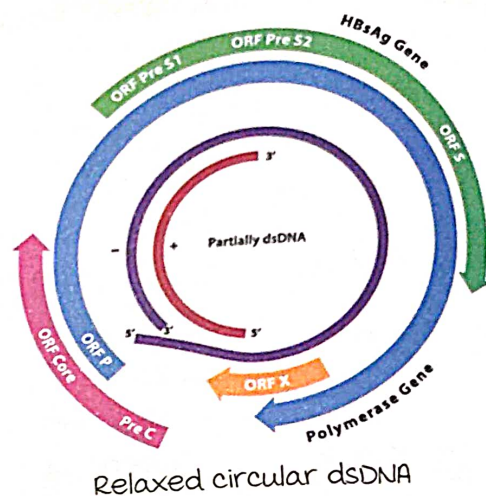
Obliterative portal venopathy

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HEPATITIS B VIRUS : PART 1

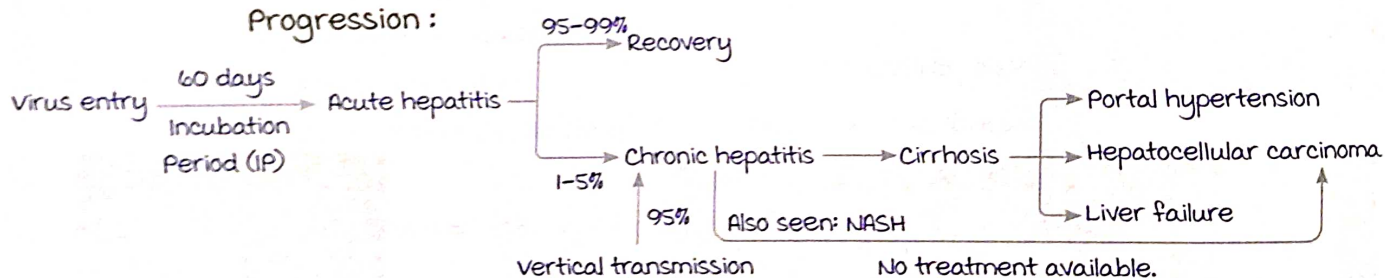
Structure :

- DNA : Hepadnavirus
- Partially double stranded : Relaxed circular dsDNA
 - Incomplete (+) strand : primer
 - Complete (-) strand : Reverse transcriptase/ polymerase
- Infective particle (virion) : 42nm Dane particle
- 10 genotypes in HBV
- 4 open reading frames (ORFs) :



Gene	Codes for	Significance
S-gene	HbsAg	Surface protein
C-gene	HbcAg	Core protein (Intracellular anti-HbcAg : in blood)
Pre C-gene (Part of C-gene)	HbeAg	Qualitative marker of infectivity & replication
P-gene	polymerase	Reverse transcriptase enzyme
X-gene	Hbx Ag	Associated with hepatocellular carcinoma (HCC)

Progression :



Infectivity :

- 100x as HIV, 10x as HCV
- 1 genotype in HAV, 4 in HEV
 - Commonest : Genotype A
 - HCC risk : Genotype C
- Found in all body fluids except stools.

Pregnancy :

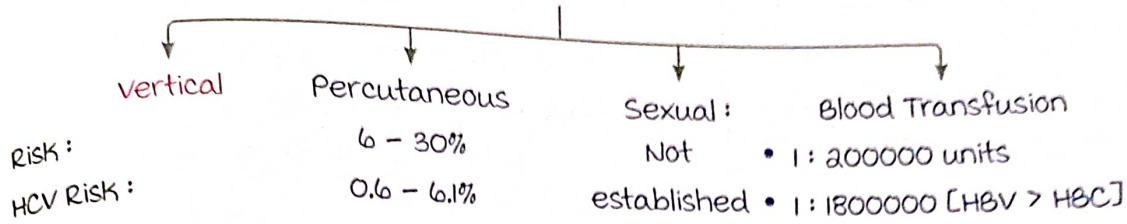
- 90/10 rule
- HbsAg +ve
 - HBeAg +ve : 90% chance of transmission
 - HBeAg -ve : 10% chance

Acute Hepatitis B

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MODE OF TRANSMISSION



No feco-oral/breast milk transmission.

Post Transfusion Hepatitis :

- Hepatitis following transfusion
- **cmv** > EBV > HCV.

SYMPTOMS

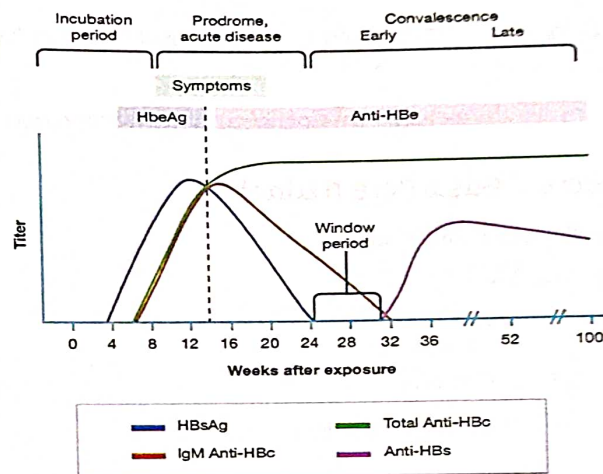
- 2-3 days of prodromal phase after 60 days of IP.
- Jaundice + ↑ ↑ enzymes (ALT/AST).

INVESTIGATIONS

- Anti-HAV IgM (m/c cause : children).
- Anti-HEV IgM (m/c cause : adults).

↓ if both -ve

- HBsAg, HBeAg, anti-HBc IgM.



HBV infection markers

Interpretations of markers :

HBsAg	+	+
HBeAg	+	-
Anti HBcIgM	+	+
Interpretation	Acute hepatitis with ↑ infectivity & replication	Low infectivity & replication/Pre core gene mutant (unable to express HBeAg)

- S gene/A gene/escape/occult gene mutation : ↓ viral load
- won't express HBsAg.
- Best test : Anti HBc IgM.
- Basal core mutant : Express 30% of HBeAg.

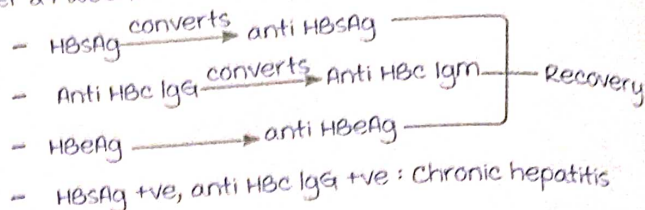
- Best marker for acute hepatitis : HBV DNA load (Quantitative marker).

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Recovery		95-99%
Chronic hepatitis		1-5%
Acute fulminant hepatitis (AFH)	Simple HBV infection	0.1-1%
	HBV + HDV confection	5-20%

FOLLOW UP

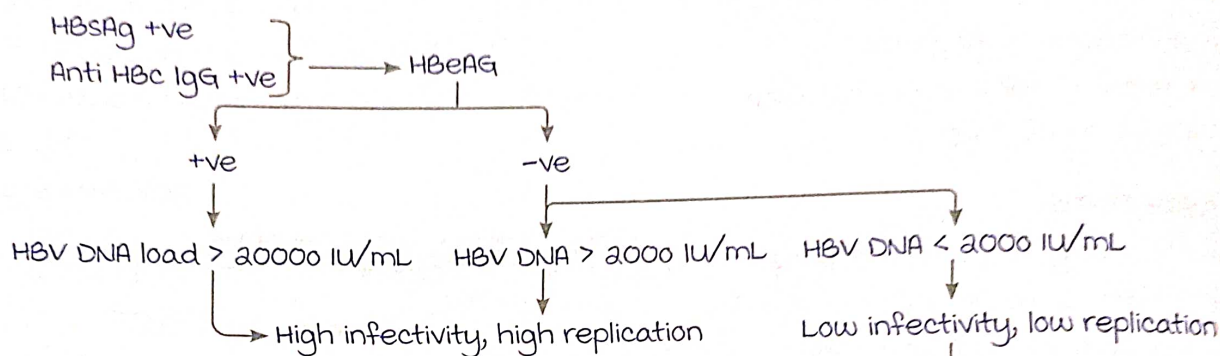
- After 24 weeks since IP/12-16 wks of symptom onset



Chronic Hepatitis

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PROFILE



Pre-Core / Basal Core mutant :

- very poor prognosis.
- ↑ risk for :
 - HCC.
 - Cirrhosis.
 - Decompensation.

SEROLOGICAL PATTERNS OF HEPATITIS B INFECTION

HBsAg	Anti-HBs	Anti-HBc	HBeAg	Anti-HBe	Interpretation
+	-	IgM	+	-	Acute hepatitis B, high infectivity
+	-	IgG	+	-	Chronic hepatitis B, high infectivity
+	-	IgG	-	-	1. Late acute or chronic hepatitis B, low infectivity 2. HBeAg negative (Precore-mutant) hepatitis B (Chronic or, rarely, acute). Do HBV DNA to confirm
+	+	+	+/-	+/-	1. HBsAg of one subtype and heterotypic anti-HBs (common) 2. Process of seroconversion from HBsAg to anti HBs (rare) (window period)
-	-	IgM	+/-	+/-	Acute reactivation of chronic hepatitis B.
-	-	IgG	-	+/-	1. Hepatitis B in remote past 2. Escape mutant
-	+	IgG	-	+/-	Recovery from hepatitis B
-	+	-	-	-	Immunization with HBsAg (After vaccination)

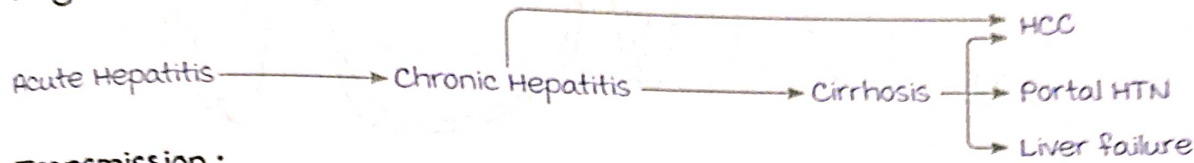
HEPATITIS B VIRUS : PART 2

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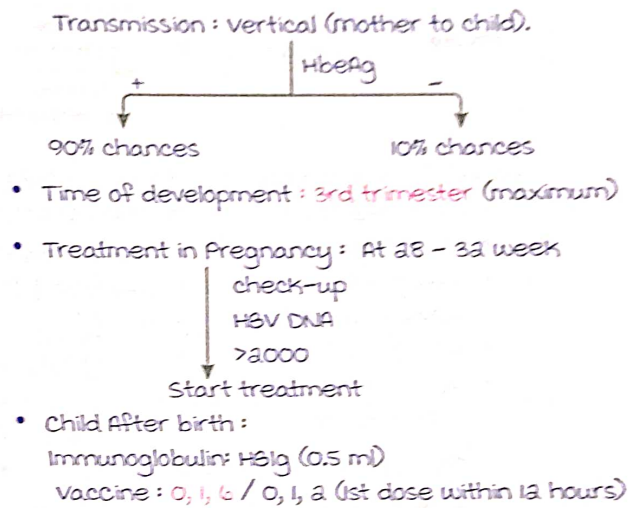
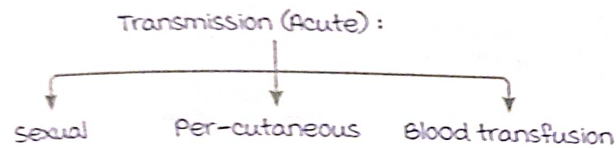
Progression, Prophylaxis and phases of Hepatitis B

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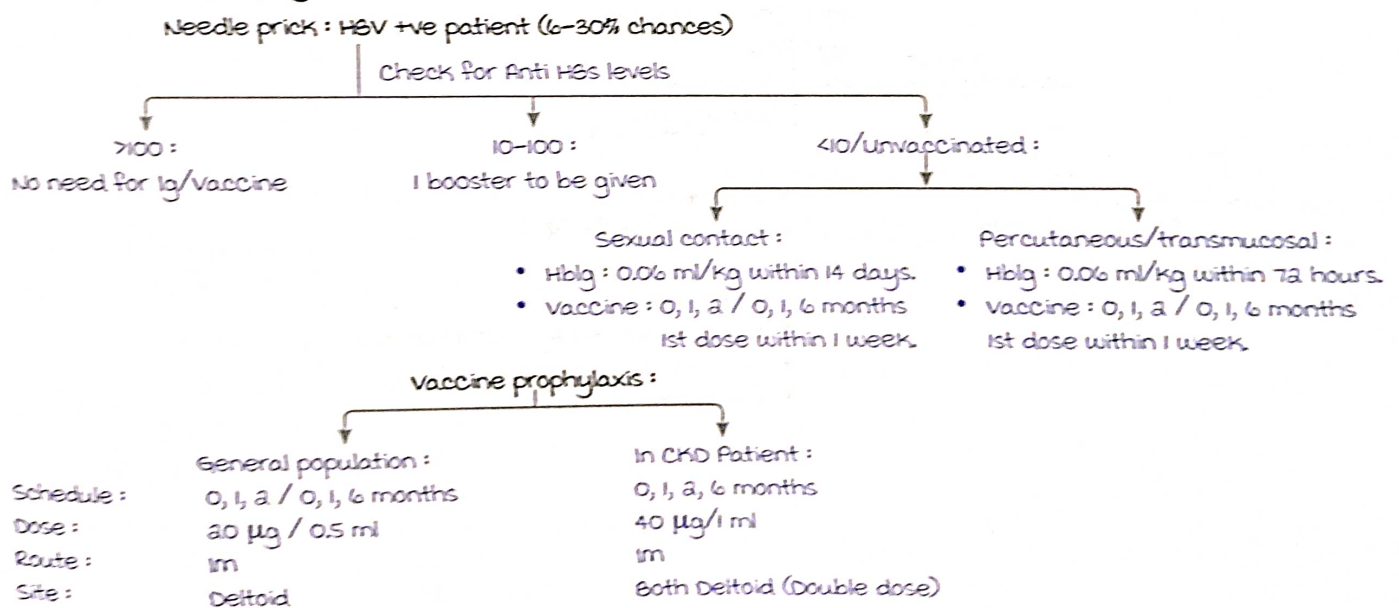
Progression :



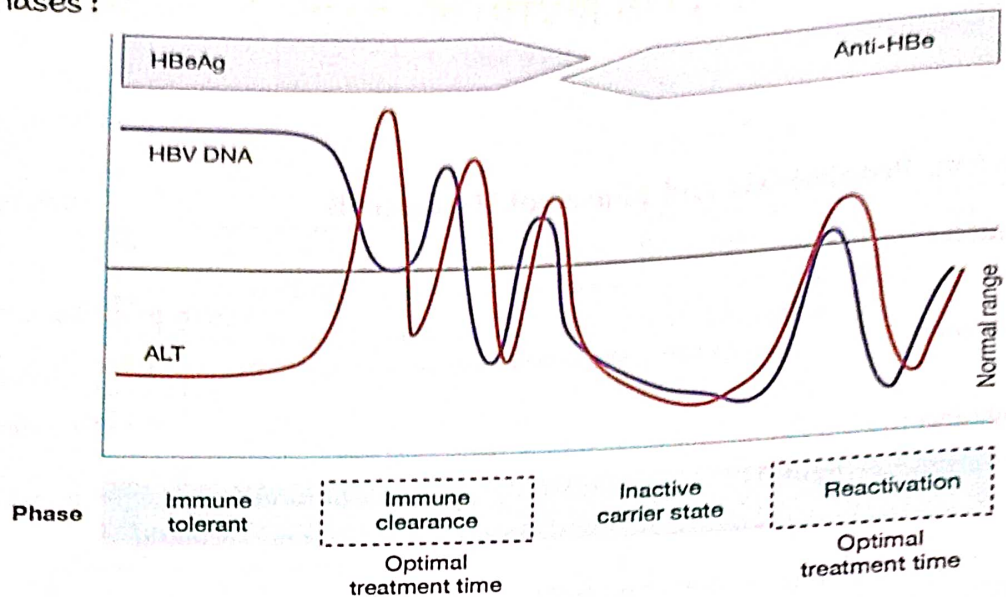
Transmission :



Post exposure prophylaxis :



----- Active space ----- Phases :



Phase I/HBeAg Positive chronic infection :

- 10 - 30 years
- HBSAg +ve
- HBeAg +ve
- HBV DNA ↑↑
- ALT : Normal → No inflammation

Phase IV/HBeAg Negative chronic hepatitis :

Presentation

Acute symptoms : Fever, malaise, vomiting

Laboratory features : ↑ S. bilirubin
↑ liver enzyme

Phase II/HBeAg Positive chronic hepatitis :

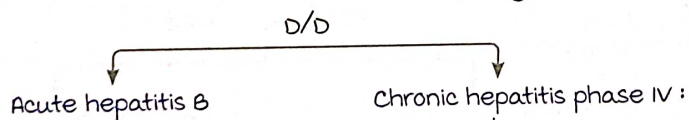
- mildly Symptomatic
- HBV DNA ↓↓
- ALT ↑↑
- HBeAg +ve

Phase III/HBeAg Negative chronic infection :

- majority population
- HBV DNA : lowest
- HBeAg - Normal

Phase V/HBSAg negative chronic infection :

- No inflammation
- minimal fibrosis



For confirmation :

- Quantitative HBS Ag (>1000 IU/ml)
- Severely Symptomatic
- Very high HBV DNA load

Seroconversion :

Conversion of HBsAg-negative patient into HBsAg-positive after Immunosuppressive therapy.

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Acute Hepatitis vs Chronic Hepatitis carrier

Both markers should be checked before immunosuppressive therapy

HBsAg -ve

Anti-HBc IgG -ve

HBsAg -ve

Anti-HBc IgG +ve

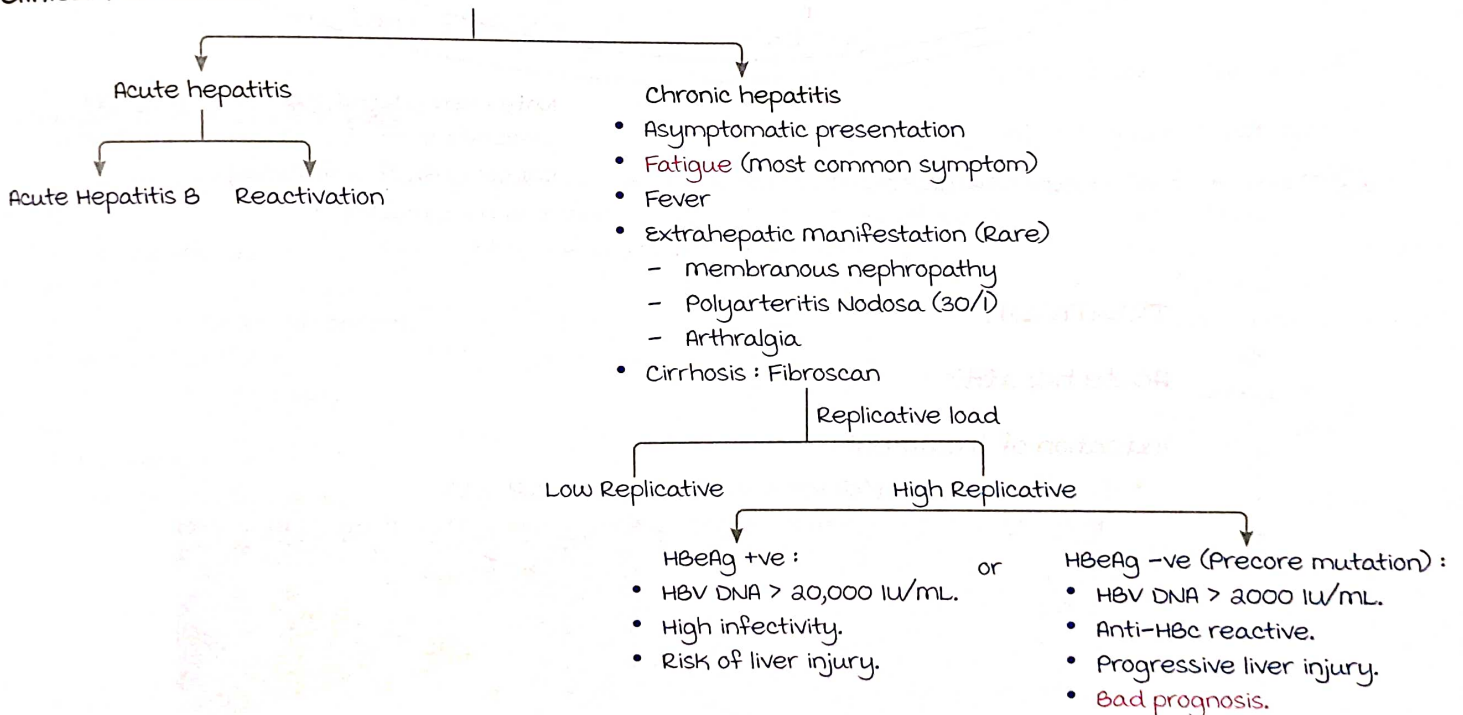
Immunosuppressive therapy

Reactivation

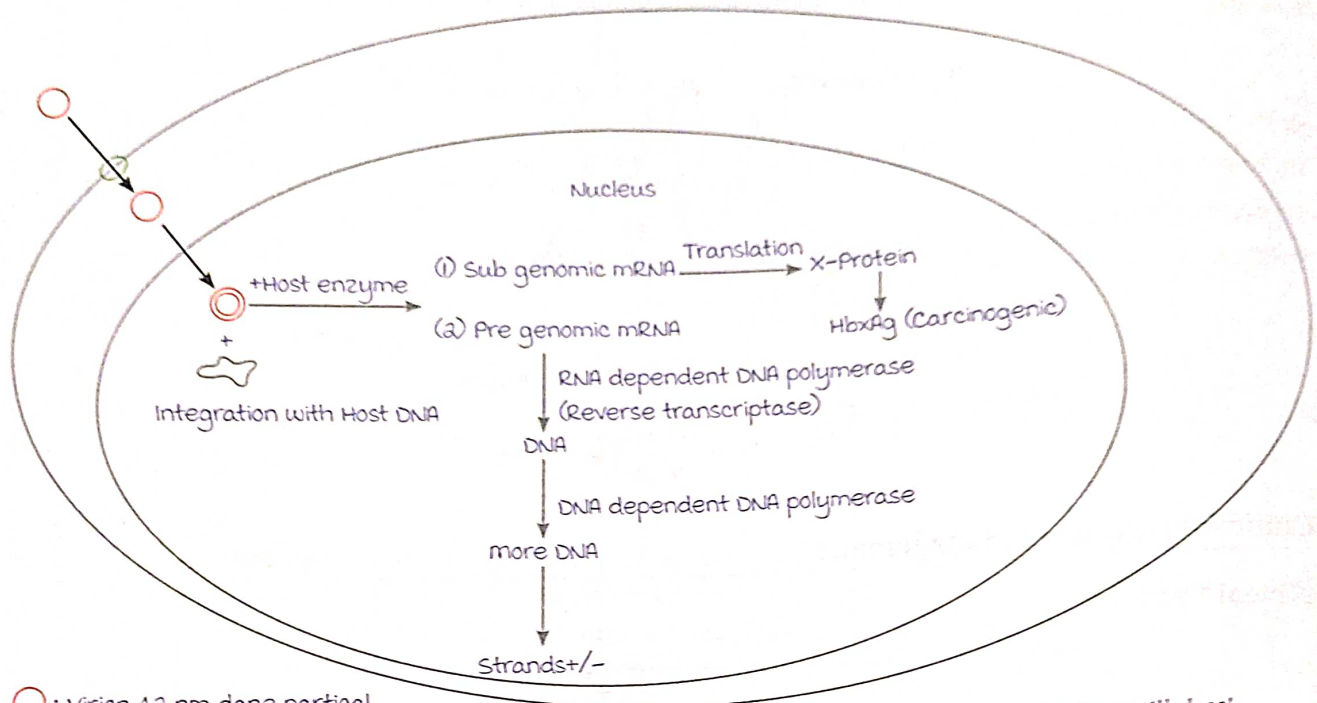
Clinical Features & Management

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Clinical Features :



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○ : Virion 42 nm dane partical

○ : Na-taurocholate receptor

○ : Covalent closed circular DNA (CCC DNA)

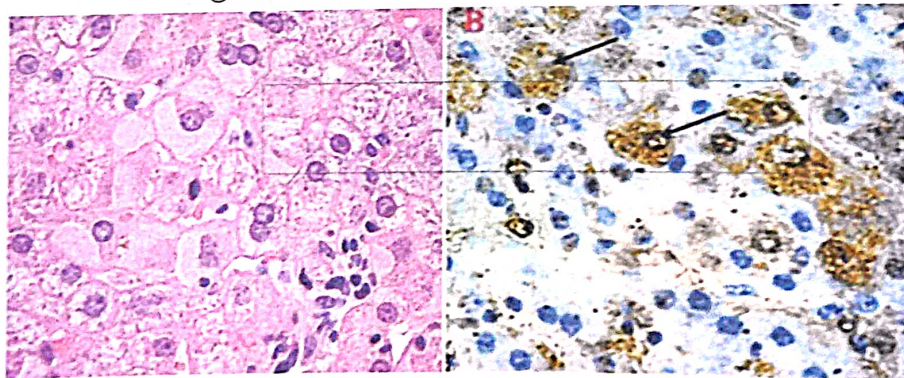
Note : HBV cccDNA integration with host genome makes it impossible to eradicate. Confers risk of malignancy.

TREATMENT

Acute hepatitis :

Indication of treatment :

- S. Bilirubin >3 mg/dl for 4 weeks.
- INR ≥ 1.5

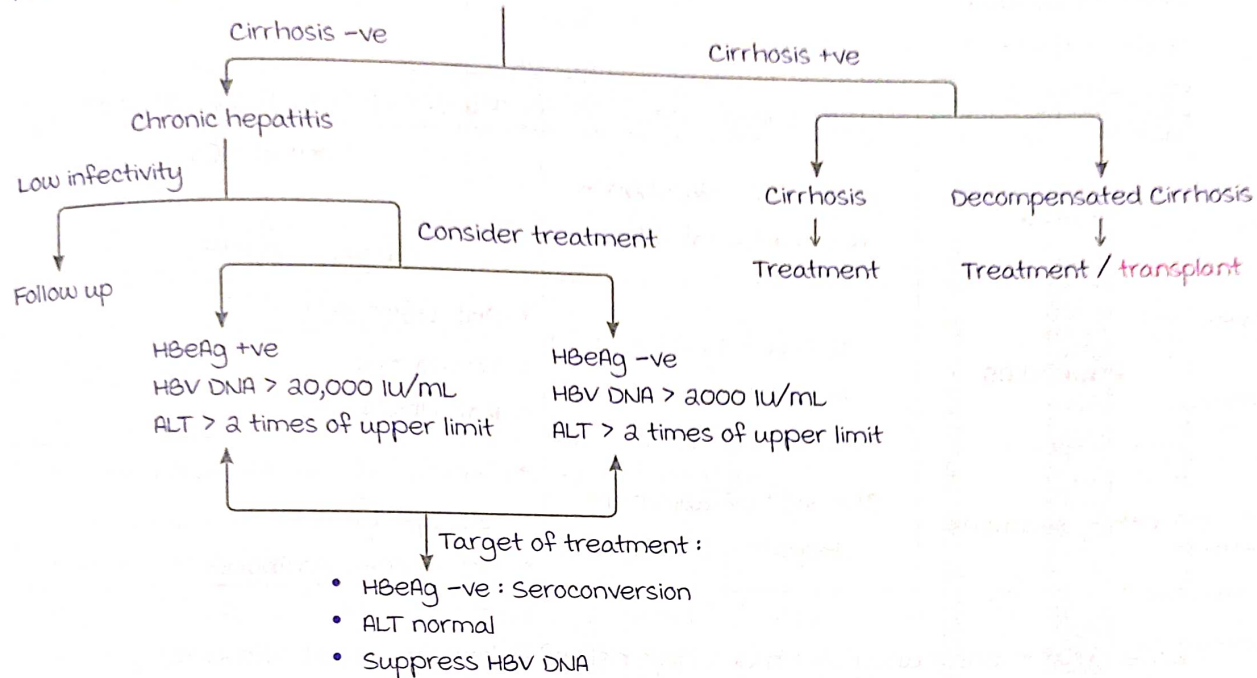


Hepatocytes with ground glass inclusions

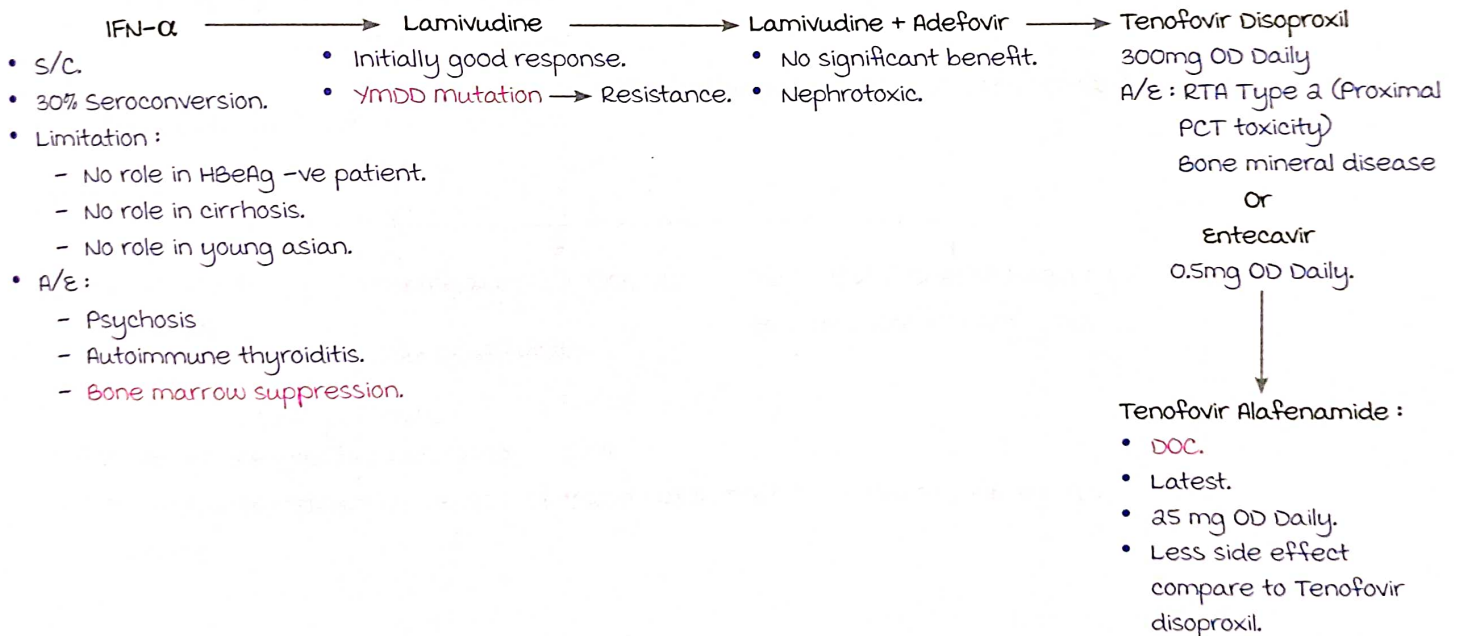
Immunohistochemical stain for HBeAg showing brownish inclusion

Chronic Hepatitis :

Initial investigation : fibroscan.



Available therapies for HBV :



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Hepatitis D Virus

Delta virus, RNA

Anti LK_m-3 Antibody

	Coinfection	Superinfection (Acute on chronic)
Cause	Acute HBV infection + Acute HDV infection	Chronic HBV + Acute Hepatitis D
Antibodies	Anti HBc IgM +ve Anti HDV IgM +ve	• Anti HBS IgG • HBSAg +ve • Anti HDV IgM
Other features	Chances of fulminant Hepatitis 5-20%	• Chronic HDV related cirrhosis (Bad prognosis) • Anti LK_m-3 Antibody

Note : HEV in pregnancy also has a high risk of fulminant hepatic failure.

HEPATITIS C VIRUS

Hepatitis C Virus (HCV)

00/01/20

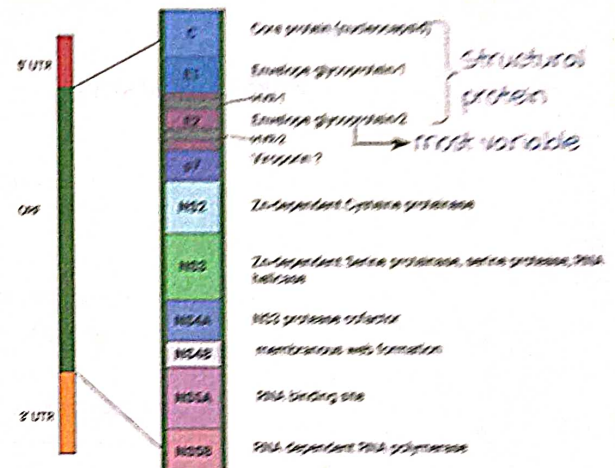
Structure :

- +ve sense RNA virus : Flaviviridae (Family), Hepaciviridae (Sub family).
 - virion (lipo viral particle) : Bound to VLDL/LDL $\xrightarrow{\text{causes}}$ **steatosis**.
 - Length : 9.7 Kb $\xrightarrow{\text{codes}}$ 3000 amino acid polypeptide.
 - Structural genes $\xrightarrow[\text{peptidase}]{\text{host signal}}$ cleaned from Non-structural genes
 - NS3 (-) : Pririr
 - NSSA (-) : Asvir
 - NSSB (-) : Uvir
- } New Directly Acting Antivirals (DAAs)
- 7 genotypes :

- Genotype 1 : m/c (World).
- Genotype 3 :
 - $\rightarrow$ m/c (India).
 - $\rightarrow$ worst prognosis.
 - $\rightarrow$ Associated with Cirrhosis/Hepatocellular carcinoma (HCC).

Note :

Hepatitis A, B, D, E do not cause steatosis.

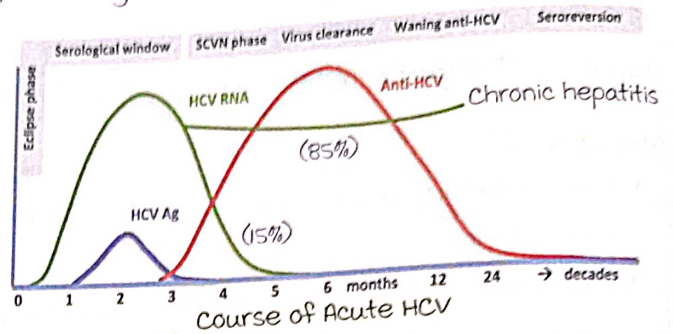
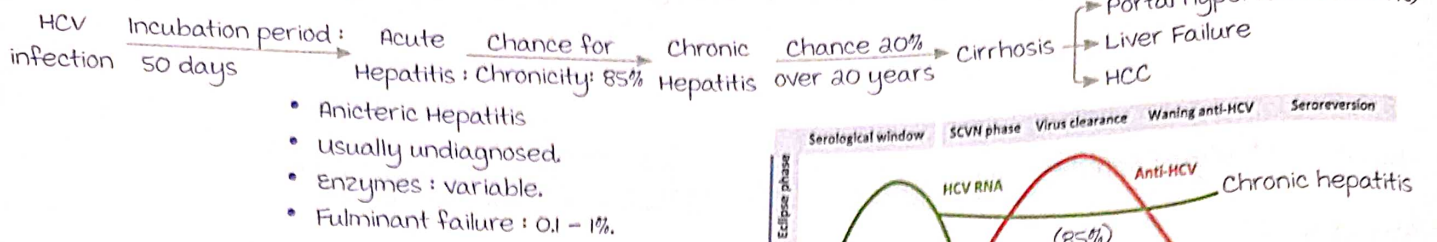


Schematic representation of HCV

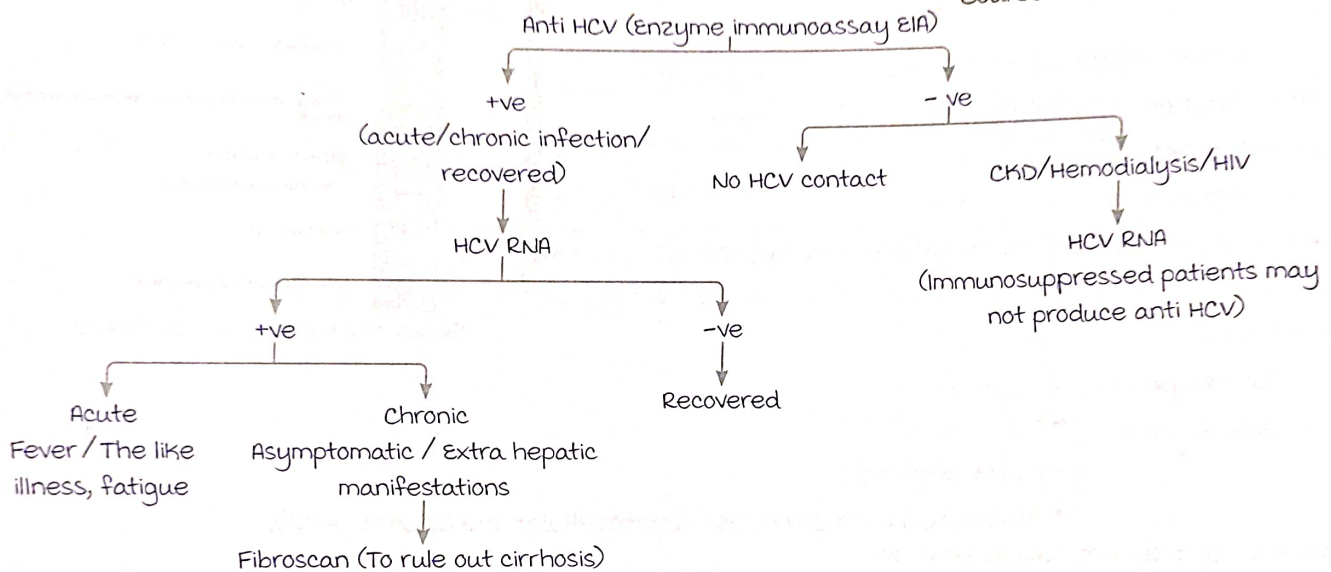
Transmission :

- Percutaneous : 0.6 - 6%
- Transfusion : 1 / 1800000
- Perinatal / Breast feeding / Sexual modes : Negligible
- m/c in :
 - IV drug users (48-90%).
 - CKD patients.
- Post exposure prophylaxis : unavailable.
- Completely cytoplasmic $\rightarrow$ no integration with host chromosome $\rightarrow$ Easy to eradicate.

----- Active space ----- Progression :



Interpretation :



Treatment :

Prognosis :

Condition	Chronic HCV	Cirrhosis	Decompensated liver disease
Sustained Viral Response (SVR)	99.5-99.7%	75	60%

Drugs :

- New DAAs, pangenomic drugs with ↑↑ SVR.
- Treat anyone > 12 years of age.
- NS3 (-) :

- Glecaprevir/simeprevir.
- Not available in India.

- NS5A (-) + NS5B (-) + Ribavirin
- velpatasvir - sofosbuvir

Combination for 12 weeks for chronic HCV

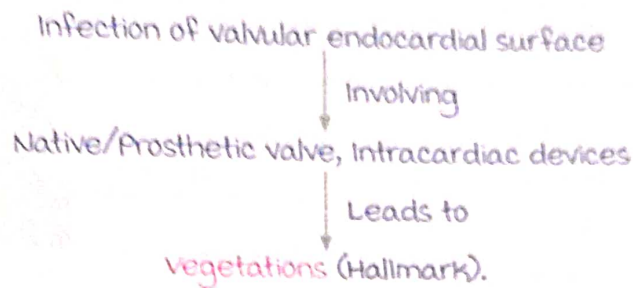
combination for 12 weeks for chronic HCV with cirrhosis

Extra hepatic manifestations :

1. Sjogren's syndrome.
 - Extranodal B cell marginal zone lymphoma (MZL) : salivary gland.
2. Essential mixed Cryoglobulinemia type 2, 3.
3. Porphyria cutanea tarda.
4. Lichen planus.
5. Classical MPGN (membranoproliferative glomerulonephritis).
6. Autoimmune thyroid illness (Hashimoto).

INFECTIVE ENDOCARDITIS

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Etiopathogenesis & Classification

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RISK FACTORS

Congenital conditions :

- Combined mitral valve prolapse (mvp) + mitral regurgitation (Highest).
- Ventricular septal defect.
- Atrial septal defect.
- Pure mvp (minimum).

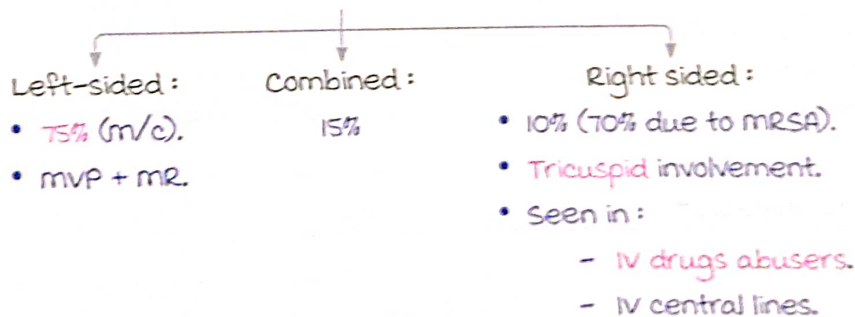
Rheumatic conditions :

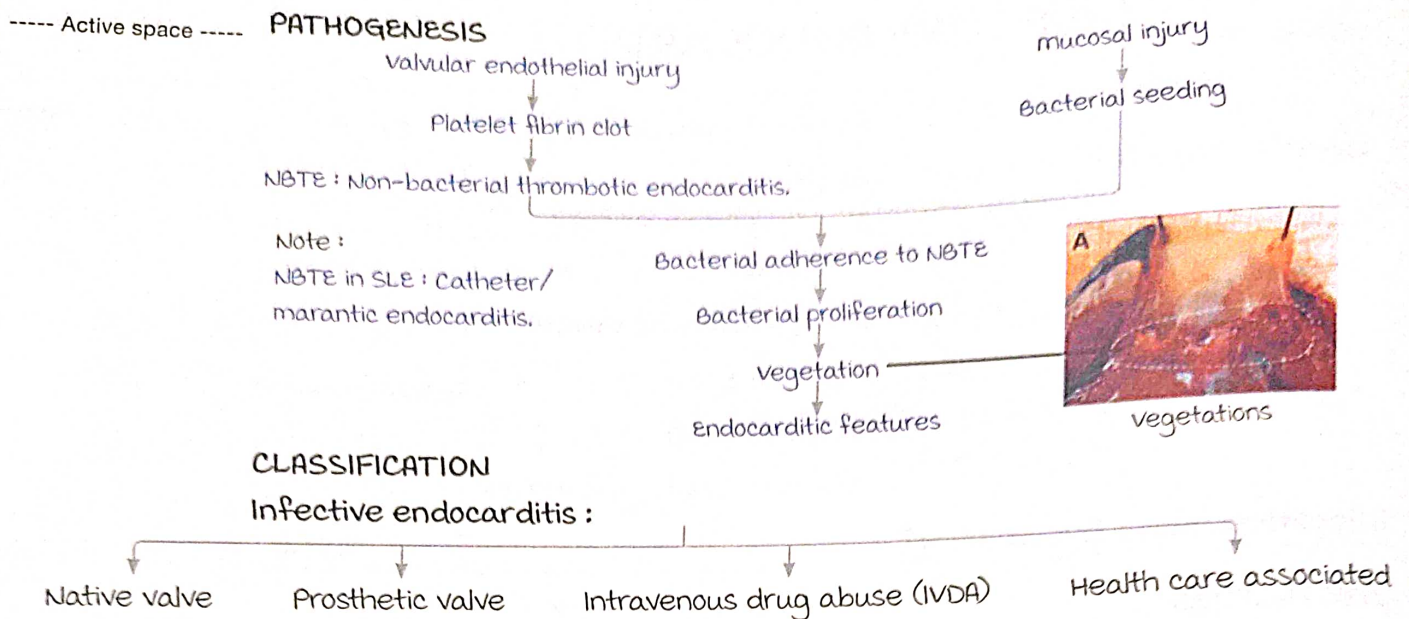
- mitral regurgitation, Aortic regurgitation.
- No risk with mitral stenosis.

Degenerative condition : Calcific degenerative aortic stenosis.

IV drug abusers.

Types :





Valvular involvement :

most common :

- valve involved : **mitral** (Left sided).
- Associated valvular heart disease :
mvp + mitral regurgitation.
- Congenital heart disease :
ventricular septal defect.

CHD associated IE : Tricuspid.

Predisposing conditions in native valves :

Predisposing conditions	15-60 years
Rheumatic heart disease	25-30
Congenital heart disease	10-20
mitral valve prolapse	10-30
Degenerative heart disease	Rare
Parenteral drug abuse	15-35
Other	10-15
None	25-45

Various endocarditis and associated organisms :

Native valve : **S. aureus** > Streptococcus viridans > Enterococci.

Prosthetic valve :

- < 12 months : **CONS** > MRSA > Gram -ve bacilli.
- > 12 months : **Streptococcus viridans** > S. aureus > Enterococci.

IV drug users associated IE : **MRSA**.

Clinical Presentation

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ACUTE IE

- very high mortality (100%).
- Associated condition : Abscess, osteomyelitis → Severe sepsis, Septic shock (MRSA)
↓
Destroyed valves.
- management : Of Sepsis.

SUBACUTE IE

H/O :

- CHD.
- Rheumatic fever in childhood.
- Treatment for RHD.



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Petechiae

Symptoms :

- Fever with chills x 2-3 months.
- Anorexia.
- Weight loss.
- Fatigue.
- Myalgia.

Signs :

- New onset cardiac murmur/worsening of an existing murmur.
- **Macroembolic events** : Stroke, ACS, retinal occlusion.
- Microembolic events : Petechiae and splinter hemorrhage.
- Splenomegaly/clubbing.
- Features of arterial embolization.

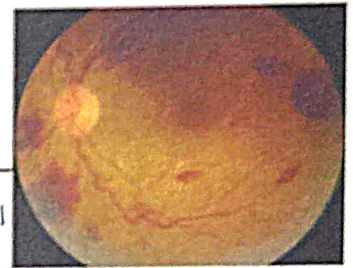
Laboratory features :

- Anemia (Normocytic).
- ↑ WBC.
- ↑ ESR.
- ↑ CRP.
- Rheumatoid factor +ve (50% cases).
- Proteinuria and microscopic hematuria (50% cases).

manifestation a/w vasculitis :

- Osler nodes
- MPGN
- RF positivity
- **Roth spots** :

- Retinal endothelial vasculitis.
- Pale centre with surrounding erythema.



	Janeway lesion	Osler node
Location	Soles, palms, thenar and hypothenar eminences, plantar surfaces of the toe. 	Finger and toe tips, pulp, thenar and hypothenar eminences. 
Size and shape	macules of variable size and irregular shape.	Nodules of 1 mm to >1 cm.
Tender	No	Yes
Course	Days to weeks	Hours to days
Type of endocarditis	Acute	Subacute
Culture	Positive, usually	Negative, usually
Histology	Septic microemboli	Vasculitis

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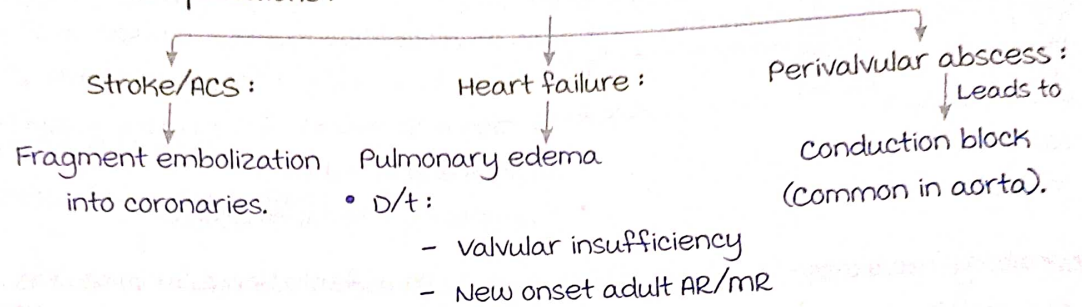
MPGN (membranoproliferative glomerulonephritis) :

Causes :

- IE ($\downarrow$ C3, $\downarrow$ C4)
- Hep C virus.
- Leprosy.
- Cryoglobulinemia.

COMPLICATIONS

Cardiac complications :



Non-cardiac complications :

- Embolic occlusion.
- Immunological.
- metastatic infection.
- mycotic aneurysm.

Right-sided IE complications :

- Pulmonary abscess.
- Empyema.
- Right heart failure.
- mycotic aneurysm of pulmonary artery.

Diagnosis

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MODIFIED DUKE CRITERIA

Criteria :

2 major
OR
1 major + 3 minor
OR
5 minor.

major criteria :

1. Blood culture :

- 3 sets (20 ml each) from 3 sites.
- Blood (10 ml aerobic + 10 ml anaerobic culture) taken over 30-60 minutes.
- Culture to be repeated every 18-24 hours.
- Positive culture :
 - Typical organism (MRSA, Strep. viridans) from 2 separate cultures.
 - Persistently positive culture.
 - HACEK group (Coxiella burnetii) single positive culture or antibodies detected.

2. Positive echocardiography :

- Oscillating intracardiac mass.
- Abscess.
- New partial dehiscence of prosthetic valve.
- New aortic/mitral regurgitation.

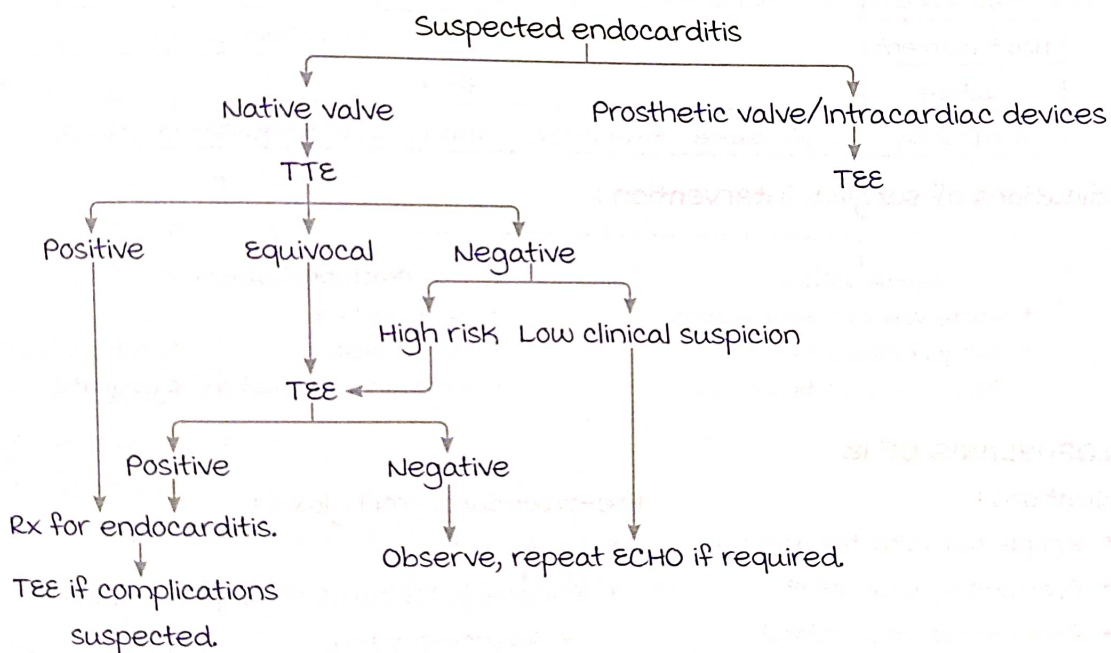
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minor criteria :

1. Predisposition to IE.
2. Fever.
3. Vascular phenomenon (Emboli).
4. Immunological phenomenon.
5. Microbiological phenomenon not meeting culture criteria.

ECHOCARDIOGRAPHY

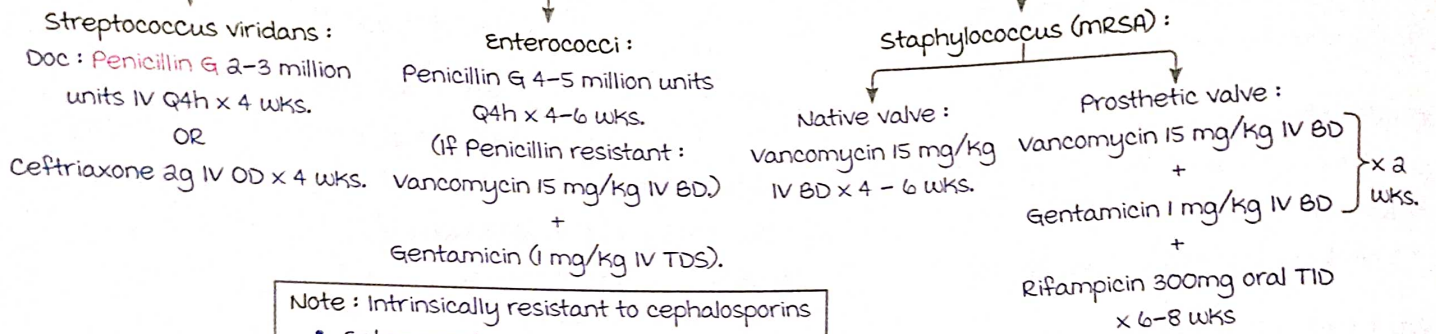
	Sensitivity (%)	Specificity (%)
TTE (Transthoracic echocardiography)	60 - 65	98
TTE (Transesophageal echocardiography)	85 - 95	85 - 98



TTE indications over TTE :

- Vegetations < 2mm.
- Myocardial abscess.
- Valve perforation.
- CIED (Cardiovascular implantable electronic device).
- Intracardiac fistula.
- Prosthetic valve.

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Treatment**Medical management :**

Note : Intrinsically resistant to cephalosporins

- Enterococci.
- Listeria.

Culture report negative v/s Culture negative :

	Culture report negative	Culture negative
Prevalence	Rare condition	m/c condition
H/o treatment	No Rx	Antibiotics
Culture	Sterile	
Medication	Ceftriaxone + Gentamicin	Vancomycin + Gentamicin x 4-6 wks

Indications of surgical intervention :

- Native valve :**
 - Acute valvular regurgitation.
 - Fungal endocarditis.
 - Perivalvular aortic abscess.
- Prosthetic valve :**
 - Heart failure.
 - Dehiscence.
 - Increasing obstruction/regurgitation.

PROPHYLAXIS OF IE**Indications :**

- People with prosthetic valve.
- Previous episode of IE.
- Post cardiac transplant.
- Unrepaired CHD.
- Repaired CHD (within 6 months).

Pre-procedural prophylaxis :

- Dental procedure.
- Invasive procedures of respiratory tract :
 - Adenoidectomy.
 - Tonsillectomy.
 - Incision/drainage of respiratory mucosa.

Medication :

- Amoxicillin 2g oral : 30-60 min before procedure.
- In penicillin allergy :
 - Azithromycin 500 mg.
 - Cephalexin 2g.
 - Clindamycin 600 mg.
- Unable to take orally :
 - Ampicillin 2g IV/Ceftriaxone 1g IV or IM.
 - In penicillin allergy : Clindamycin 600 mg IV.

TROPICAL INFECTIONS : SYNOPSIS

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Fever in Infections

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Fever on day 1 & 2 :

Treatment : Paracetamol 650mg QID + Decongestants.

screening : Look for foci of infection.

Fever on day 3 :

Pattern of fever :

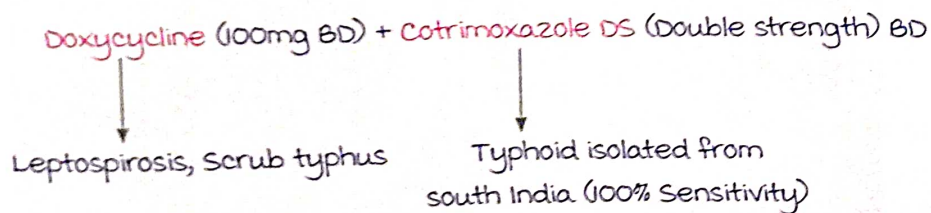
- Salmonella : Step ladder/Rising pattern.
- malaria : Chills and rigors a/w fatigue.
- Leptospirosis, scrub typhus : A/w myalgia (Calf muscle), conjunctival suffusion in a farmer.
- Dengue :
 - High grade fever a/w :
 - a. Retro-orbital pain.
 - b. Occipital pain.
 - c. vomiting (1-2 episodes).

Investigations :

- CBC profile.
- Dengue : **NS-I antigen**.
- malaria : **Quantitative buffy coat (QBC)**.
 - Repeated every 12 hours.
 - Ruled out : If 3 consecutive tests negative.

Fever on day 4 :

Start on :



Fever on day 5 :

Blood culture : 10mL each for anaerobic, aerobic and fungal cultures.

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Salmonellosis & Dengue**SALMONELLOSIS**

Clinical presentation :

Week	Features
Week 1	• Step ladder fever $\pm$ chills. • Relative bradycardia.
Week 2	• Vague abdominal pain, GI disturbances. • Hepatomegaly, splenomegaly, rose spots.
Week 3	• Complications : - Encephalopathy. - Ileocolic hyperplasia $\rightarrow$ GI perforation, bleeding.

Salient features :

Infective dose : 10^5 organisms/mL.

Complication : Gall stones in chronic carrier state (6%).

Incubation period : 5-20 days.



Rose spots

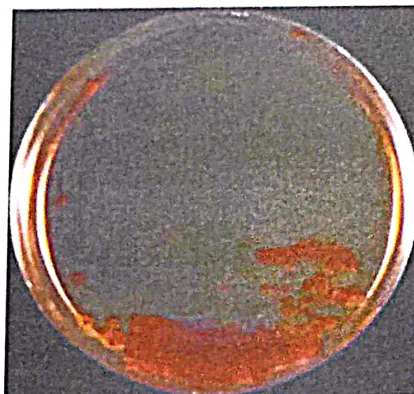
Investigations :

Widal :

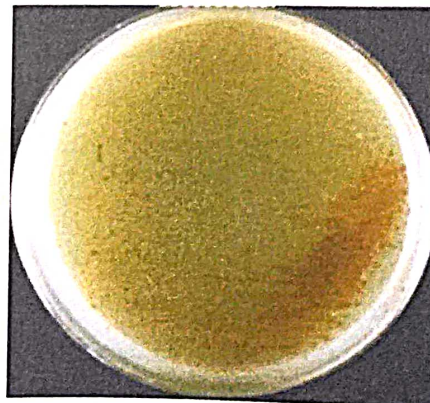
- Low positivity (1st and 2nd week).
- 80-100% positivity (3rd week).

Blood culture :

- 90% positivity (1st week).
- Culture media.



macConkey agar

Wilson & Blair bismuth sulphite agar :
Jet black colonies

Pathogenesis :

Transmission : Feco-oral route.

Salmonella invade the small intestine

via blood stream (WBC)

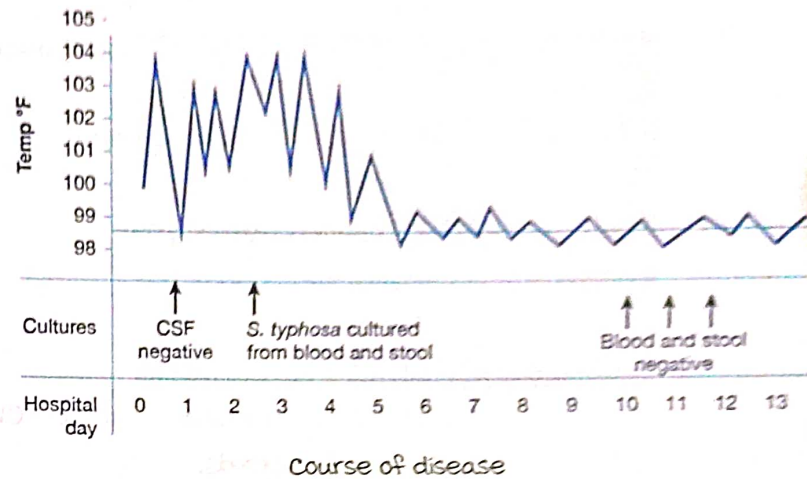
In liver, spleen & bone marrow

multiply & re-enter

Invades & multiplies in

- Gallbladder.
- Biliary system.
- Lymphatic system of bowel

Pass into intestinal tract and causes ileocolic hyperplasia.

**Treatment :**

Uncomplicated cases :
 Cotrimoxazole (DS) BD x 14 days.
 OR
 Azithromycin 1gm OD x 7 days

Complicated cases :
 Ceftriaxone 2gm I/V x 14 days

DENQUE**Incubation period :**

7-10 days.

Disease course :

Day 1 to 3 : NS-I antigen positive.

Symptoms :

- High grade fever.
- Headache.
- Retro-orbital pain.
- Vomiting.

Day 4 onwards :

NS-I antigen negative

- Fever subsides.
- ↓ Platelet counts (Not <50,000)
- Rash.
- Leucopenia.

Dengue without warning signs

----- Active space -----

Day 4-7 : (most crucial period)

In some patients

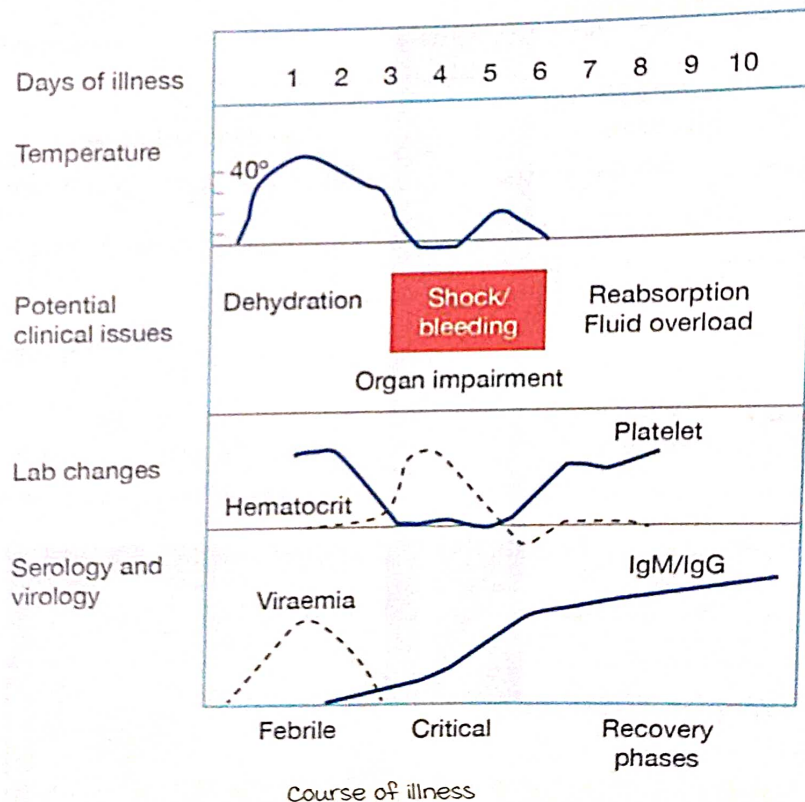
- ↓ Platelet count (Bleeding).
- Capillary leak syndrome (Pleural/pericardial effusion, ascites).

Dengue with warning signs/organ failure

management

monitor and maintenance :

- BP by I/V fluids.
- IVC diameter.
- Hematocrit.
- Platelets (Administer if $<10,000$ OR $<20,000$ + bleeding).
- No use of steroids.



Note :

Igm become positive : Day 6 onwards.

Leptospirosis & Scrub typhus

Transmitting agent : Rodents.

Incubation period : 10 days.

Clinical presentation :

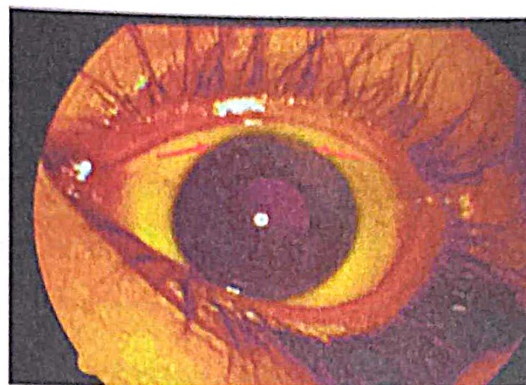
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----- Active space -----

1st week

Features :

- Abrupt onset fever (Low to intermediate grade).
- myalgia.
- Calf muscle tenderness.
- Conjunctival congestion.



Conjunctival suffusion

7-14 days (vital period)

a. Other complications :

- Hepatorenal syndrome :

-Hepatitis with jaundice + Acute tubular necrosis.

- Thrombocytopenia.

Non-oliguric AKI with :

1. Hypokalemia.
2. Hypomagnesemia.

b. Immunological manifestations :

- Myocarditis.
- Secondary MR/AR.
- Aseptic meningitis.

- Acute respiratory distress syndrome.
- Rhabdomyolysis.

Diagnosis :

Criteria : **modified Faine's criteria**.

Gold standard test : microscopic agglutination test (MAT).

Lepto Igm ELISA : Positive 7 days onwards.



MAT

Treatment :

Doxycycline + Inj. Ceftriaxone 2g I/V OD.

----- Active space ----- **SCRUB TYPHUS**

Infectious agent : *Orientia tsutsugamushi*.

Disease progression :

- Endothelial proliferation (Loss of endothelial integrity).



micro infarct in kidney
(ATN + Thrombotic
microangiopathic pattern)

- No immunological phase.

Clinical features :

- Similar presentation as leptospirosis.
- Characteristic feature : *Eschar*.
- Lymphadenopathy at the end of one week.
- If untreated → CNS involvement
(within one week) :
 - Sensorineural deafness (Reversible).
 - Opsoclonus, myoclonus.



Eschar

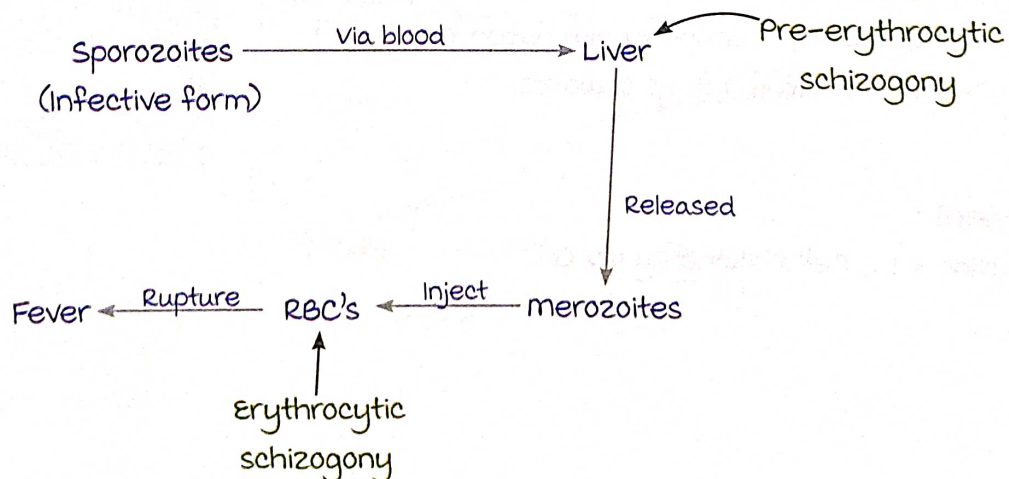
Treatment :

- Doxycycline 100mg BD x 7 days or ;
- Azithromycin 500mg OD x 5 days

Malaria

00:30:55

Life cycle :



Clinical features :

- High grade fever with chills and rigors.
- Extreme fatigue

Investigation :

Quantitative buffy coat (QBC).

----- Active space -----

Complication :

sequestration of infected RBC's in vital organs → Organ ischemia.

Miscellaneous :

- On RBC, parasite bind to **glycophorin A** receptor.
- Duffy negative RBCs are protective against *P. vivax*.
- sequestration of RBCs inside vital organs d/t :
 - Cyto-adherence, rosetting, agglutination of RBCs.
- **sickle cell trait** and **G-6PD deficiency** protects against *P. falciparum*.
- major organ involved in trapping RBCs : Spleen.

Characteristics of plasmodium species :

Characteristic	<i>P. falciparum</i>	<i>P. vivax</i>	<i>P. ovale</i>	<i>P. malariae</i>
Duration of erythrocytic cycle (Hours)	48	48	50	72
Red cell preference	Younger cells	Reticulocytes and cells upto 2 wks old.	Reticulocytes	Older cells.
Pigment color	Black	Yellow-brown	Dark-brown	Brown-black
Ability to cause relapses	No	Yes	Yes	No

Uncomplicated malaria : (D/t *P. falciparum*/*P. vivax*)

- Fever with chills/ fatigue/ diaphoresis.
- Tachypnea/ tachycardia/ splenomegaly.
- myalgia (Not severe).

Complicated malaria : (D/t *P. falciparum*)

- Anemia.
- Acidosis (Anaerobic metabolism → Lactic acidosis).
- Renal failure (Sequestration of RBCs).
- Non-cardiogenic pulmonary edema (ARDS).
- Jaundice

- Cerebral malaria (Convulsions, seizures).
- Hypoglycemia.
- Hypotension/shock.
- Hemoglobinuria (Black water fever).

----- Active space -----

Diagnosis :

microscopy (Day 3) :

Thick smear :
Gold standard.
Giemsa staining and
light microscopy.

Thin smear :
Not sensitive

QBC test :

- Stain : Acridine orange.
- 3 QBC tests 12 hours apart $\xrightarrow{\text{Negative}}$ No malaria.

Rapid diagnostic test : HRP2 (Not validated).

Treatment :

Uncomplicated malaria : (Oral therapy)

- Artemether 20mg + Lumefantrine 120mg (First line).
- 4 tablets BD x 3 days.
- Atovaquone + Proguanil.
- Clindamycin + mefloquine.
- Doxycycline + Quinine.
- Artesunate + Amodiaquine.
- Artesunate + mefloquine.

Complicated : (I/V therapy)

- Day 1 : 3 doses I/V Artesunate (120mg/2.4mg per kg) at 0, 12 and 24 hours.
- Day 2 : 2.4 mg/kg single dose.
- Day 3 : 2.4 mg/kg single dose.

HIV

1061

----- Active space -----

History :

- First reported case : 1981.
- AIDS (Acquired immunodeficiency syndrome) term coined by CDC : 1982.
- 1st isolation of virus : 1983 → By Luc Montagnier and colleagues → Named as Lymphadenopathy Associated Virus (LAV).
- World Aids day : 1 December (Since 1987).

CDC Staging :

Stage	CD4 count	CD4 %	Clinical evidence
Stage 0	Early HIV infection.		
Stage 1	≥ 500 cells/mm ³	≥ 26	No AIDS-defining condition.
Stage 2	200-499 cells/mm ³	14-25	No AIDS-defining condition.
Stage 3	< 200 cells/mm ³	< 14	Documentation of AIDS-defining conditions.

AIDS Defining Illnesses

00:11:00

1. Malignancies :

- Lymphomas :
 - Burkitt's lymphoma.
 - Immunoblastic lymphoma.
 - Primary CNS lymphoma.
- Invasive cervical cancer.
- Kaposi's sarcoma.

EBV mediated.

2. Viral infections :

- Cytomegalovirus (Other than liver, spleen and lymph node involvement).
- Herpes simplex (Except zoster).
- JC virus : Progressive multifocal leukoencephalopathy.

3. Fungal infections :

- Candidiasis of bronchi/trachea/lungs/oesophagus.
- Cryptococcosis (Extrapulmonary).
- Coccidioidomycosis (Disseminated or extrapulmonary).
- Histoplasmosis (Disseminated or extrapulmonary).
- Pneumocystis jirovecii.

----- Active space -----

4. Parasites :

- Cryptosporidiosis (Chronic intestinal >1 month duration).
- Isosporiasis (Disseminated or extrapulmonary).
- Toxoplasmosis of brain.

5. Bacterial infections :

- mycobacterium avium complex.
- mycobacterium tuberculosis.
- Salmonella septicemia.

CATEGORY B

Not AIDS defining illness :

- Vulvovaginal candidiasis.
- Oropharyngeal candidiasis (Thrush).
- Pelvic inflammatory disease.
- Cervical dysplasia/Carcinoma in-situ.
- Fever ($>38.5^{\circ}\text{C}$) or diarrhea lasting >1 month.
- Idiopathic thrombocytopenic purpura.
- Herpes zoster (Shingles).
- Peripheral neuropathy.
- Hairy leukoplakia.

Features of HIV

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HIV in India :

First case detected (Chennai) : 1986.

STRUCTURE**Retroviridae family :**

Delta virus : HTLV-I.

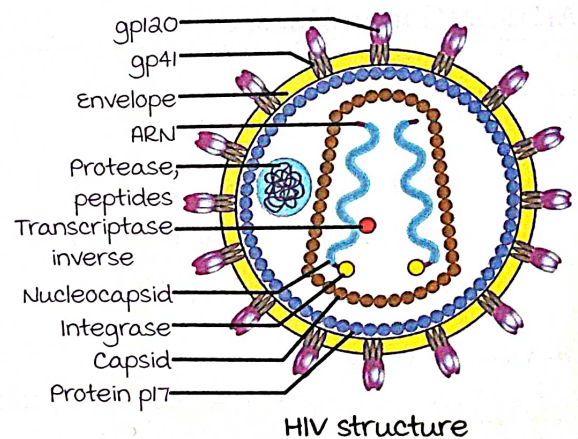
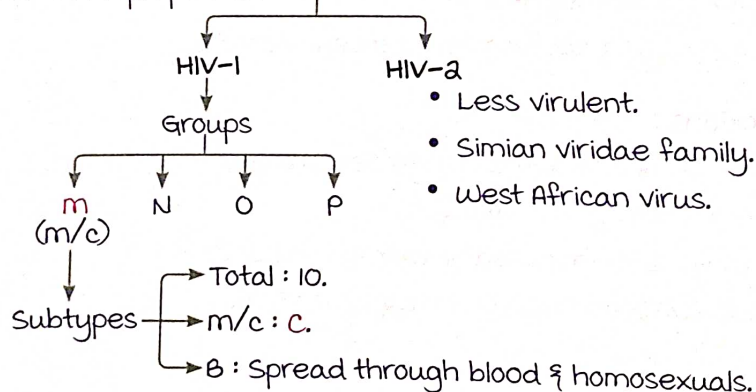
- Adult T-cell lymphoma/leukemia.
- Tropical spastic paraparesis.

Lentivirus

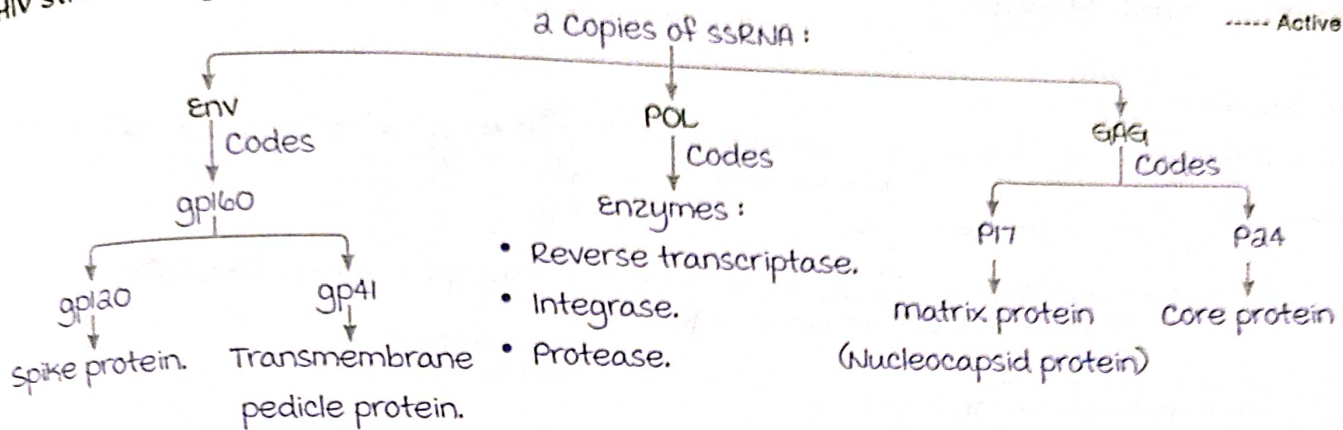
- HIV-1.
- HIV-2.

Lentivirus serotypes :

On the basis of envelope protein :



HIV structural genes :



HIV non structural genes :

- Nef/P₅₅ : Negative effector → Downregulates host CD4 cells.
- vif : Viral infectivity factor.
- Rev : Regulation of viral expression.
- Tat : Transcription activator.
- vpr : Viral protein R.
- vpu/vpx

INFECTION SOURCE

High concentration	Low concentration
• Blood. • Semen/Vaginal fluid (As high as blood). • Pus from sores. • CSF.	• Sweat. • Tears. • Urine. • Saliva (SLIP : Salivary lymphocyte inhibitory protein) • Breast milk.

MODES OF TRANSMISSION

Sexual > Parent to child > IV drug abuse > Blood transfusion > Needle stick.

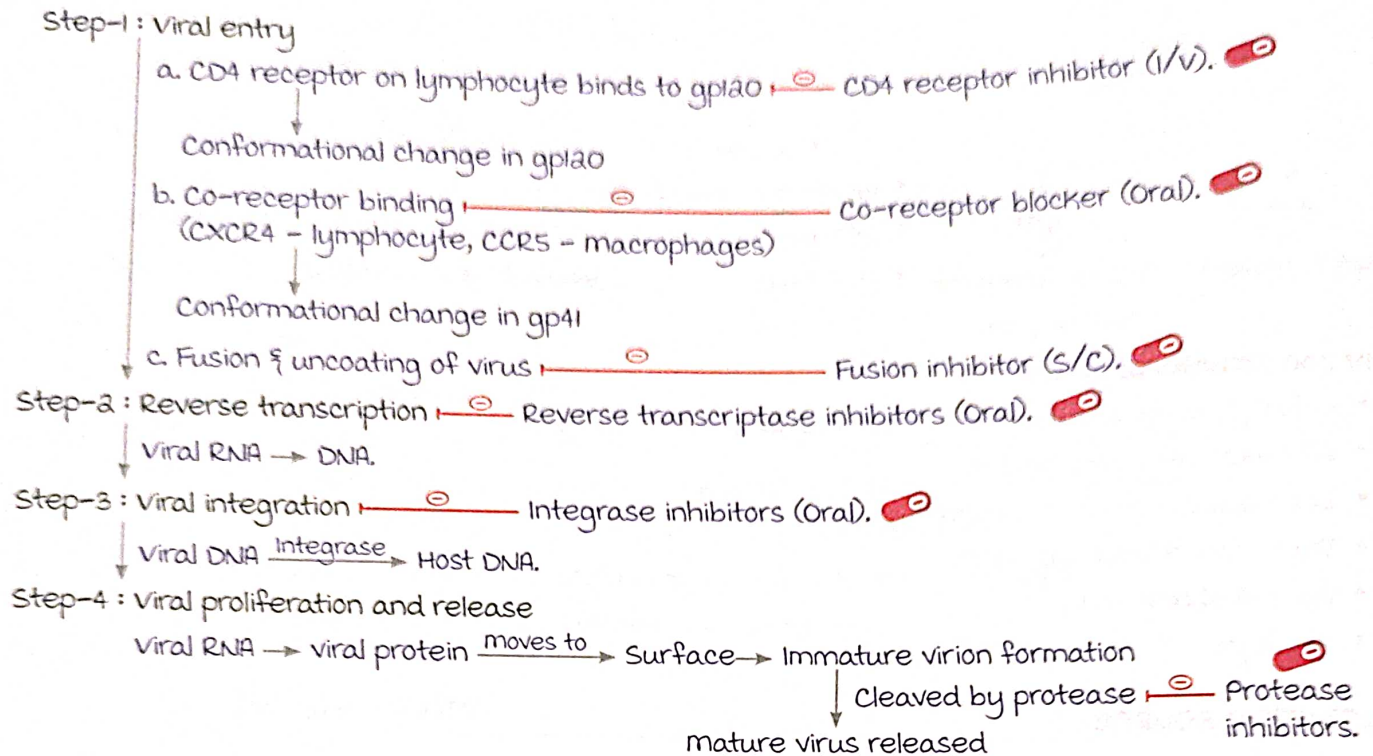
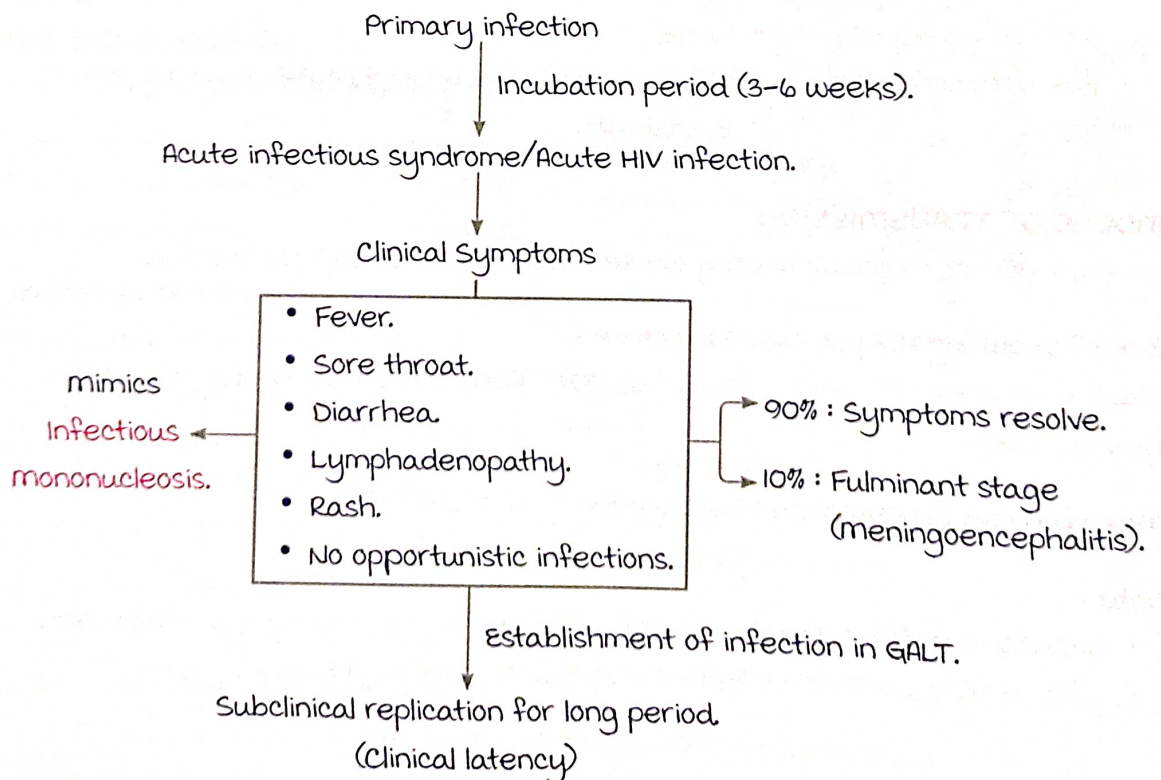
Risk of acquiring HIV per 10,000 exposure :

maximum : Blood transfusion (9250/10,000) > Parent to child > IV drug abuse > Needle stick.

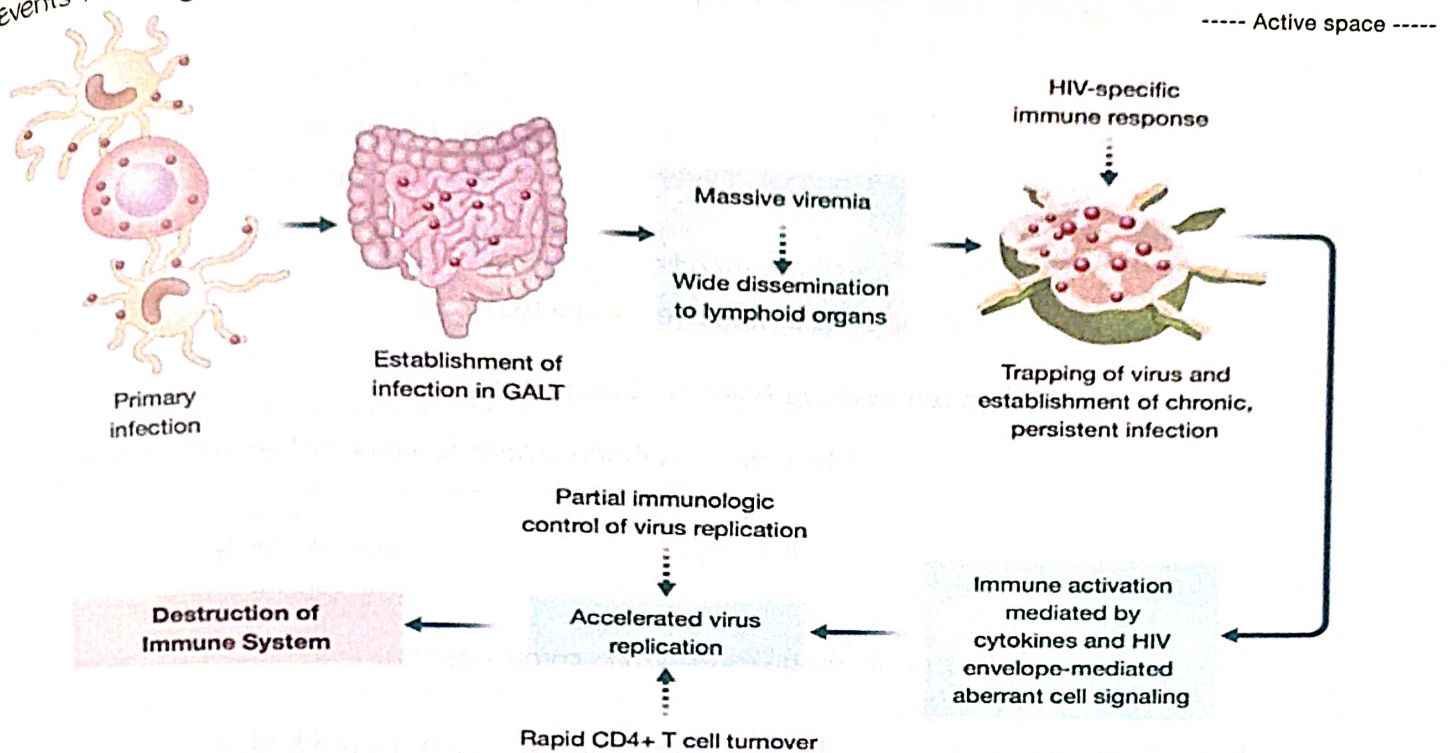
maximum risk in pregnancy : During delivery (20-30%).

Note :

- Delta32 mutation : mutation in CCR5 receptor on macrophage → Resistance.
- DC sign : C type lectin receptors on dendritic cells bind to HIV.

----- Active space ----- **Viral Replication****Disease Progression**

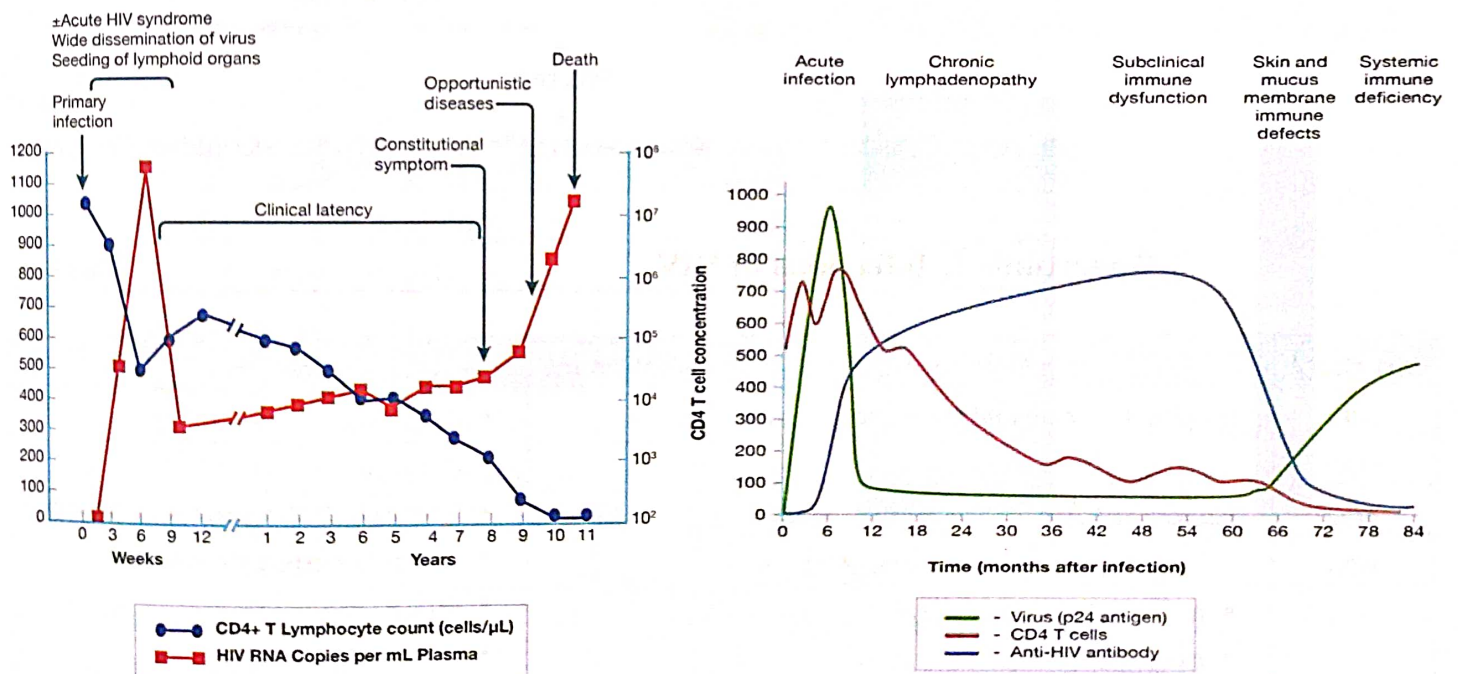
events following primary HIV infection :



Diagnosis

00:33:10

Course of untreated HIV infection :



Detection of HIV components :

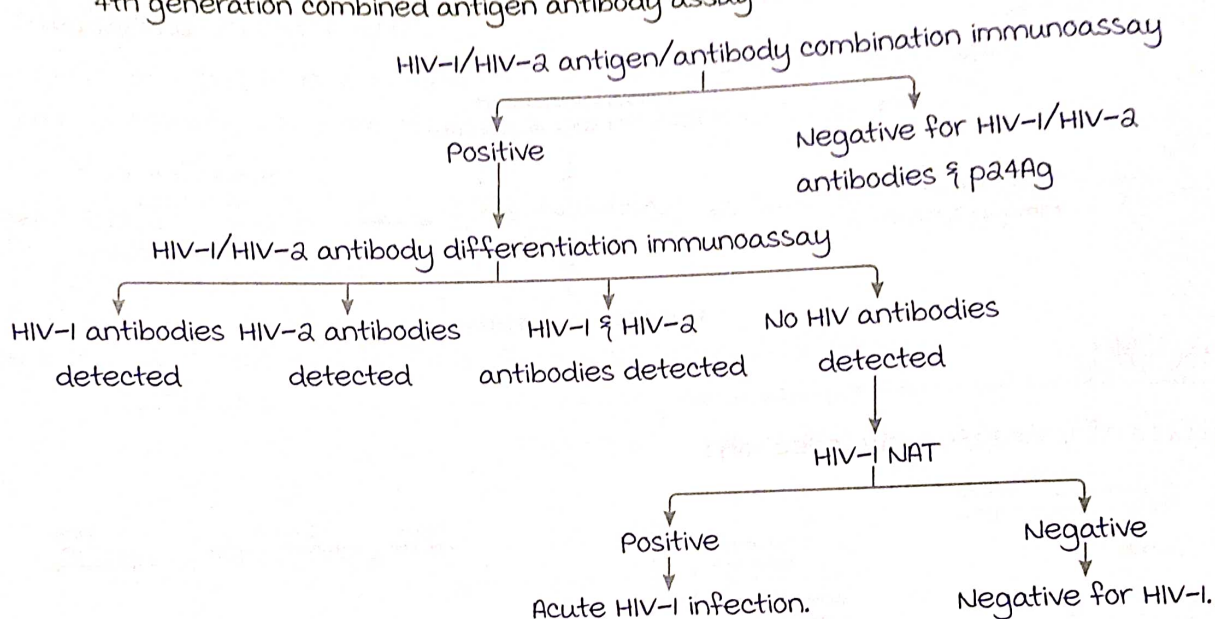
- Antibodies in circulation : 3-6 weeks (Following primary infection).
- HIV RNA : 12th day/1-2 weeks (Earliest).
- p24 levels : 16th day/2-3 weeks.

----- Active space ----- **4th generation combined antigen antibody assay :**
Currently in use.

Interpretation :

- If negative : No HIV.
- If positive :
 - Look for specific antibodies against HIV 1/2
 - If antibodies are negative → HIV RNA load.

4th generation combined antigen antibody assay :



Opportunistic Infections in HIV

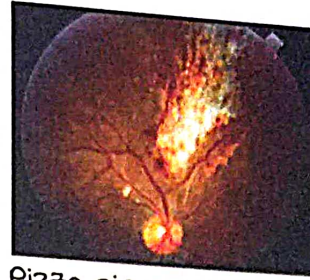
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CD4 count	Infection
600	Lymphadenopathy/thrombocytopenia
500	H2V
400	Kaposi sarcoma (At any CD4 count)
300	Oral hairy cell leukemia/TB
< 200	• mucocutaneous herpes • Cryptosporidium • microsporidium • Pneumocystis jirovecii • miliary TB • Esophageal candidiasis

CD4 count	Infection
< 100	• Cerebral toxoplasmosis • Cryptococcal meningitis • Primary CNS lymphoma (50-100) • HIV associated dementia • PMLE
< 50	• CMV retinitis • Mycobacterium avium intracellulare disease.

CMV:

- **Retinitis** > Colitis > Esophagitis > Pneumonitis.
- Painless progressive loss of vision.
- Asymmetric B/L involvement with perivascular hemorrhage/white fluffy exudate.
- Treatment: IV Ganciclovir x 3 weeks.



Pizza pie appearance

----- Active space -----

MAC:

- Average CD4 at diagnosis: <10.
- Highly disseminated disease with:
 - Fever
 - Night sweats.
 - weight loss
 - Lymphadenopathy.
- Pulmonary involvement is rare.
- Treatment: **Clarithromycin + Ethambutol**.
- Prophylaxis (CD4 <50): Azithromycin.

Treatment

00:44:32

All HIV must be treated.

DRUGS

Class	Drugs
CD4 receptor inhibitors	Ibalizumab (IV).
Coreceptor inhibitors	maraviroc (Oral).
Fusion inhibitors	Enfuvirtide (s/c).
Integrase inhibitors	Dolutegravir, Raltegravir.
Protease inhibitors	Ritonavir, Nelfinavir, Atazanavir.
Reverse transcriptase inhibitors	
1. Nucleoside reverse transcriptase inhibitors/NRTI	Abacavir, Lamivudine, Emtricitabine.
2. Non-nucleoside reverse transcriptase inhibitors/NNRTI	Nevirapine, Efavirenz, Delavirdine.
3. Nucleotide reverse transcriptase inhibitors	Tenofovir, Disoproxil.

----- Active space -----

REGIMENS**3 drug combination :**

Preferred combination :

Nucleotide reverse transcriptase inhibitors + NRTI + Integrase inhibitor.

TED regimen :

Tenofovir disoproxil + Emtricitabine + Dolutegravir.
(300 mg OD) (200 mg OD) (50 mg OD)

NACO regimen/ TLD regimen :

Tenofovir disoproxil + Lamivudine + Delavirdine.
(300 mg OD) (300 mg) (50 mg OD)

Needle stick injury : TLD regimen single pill HS.

Prophylaxis (After needle injury) :

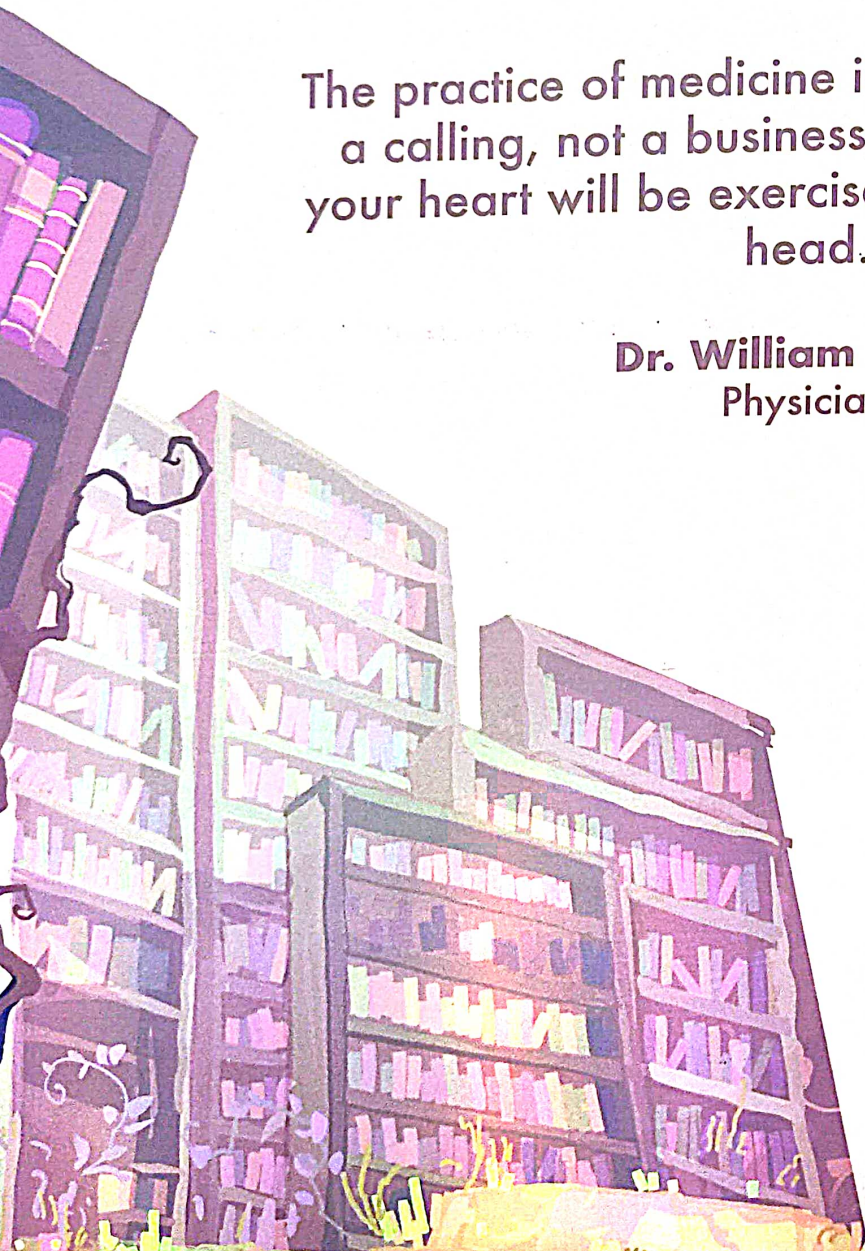
- Optimally effective if started within 72 hours.
- TED regimen for 4 weeks.

S/E : Bone toxicity with tenofovir.

“

The practice of medicine is an art, not a trade;
a calling, not a business; a calling in which
your heart will be exercised equally with your
head.

Dr. William Osler,
Physician



MARROW